MMA3001 Project: *A Reproducible Surrogate-Modelling Workflow for a Composite Beam Benchmark*

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Abstract

This project develops a reproducible workflow for learning a structural response model from finite-element analysis (FEA), calibrating an effective section stiffness from a small number of known reference observations, and checking predictions on held-out geometries. The method is first exercised on a straight, linearly width-tapered cantilever with an analytical small-deflection reference before application to a curved composite leaf-spring landing gear. The current PyAnsys input specifies a 120-case, 10 N static laminate sweep over geometry, thickness, and effective stiffness. A full quadratic response surface models FEA compliance; bounded least squares estimates the effective stiffness scale directly from analytical observations. Held-out cases test transfer to unseen geometries at the same thickness and fitted stiffness. A 120-case, 10 N PyMAPDL SHELL181 sweep and quadratic fit have been evaluated with the revised direct effective-stiffness calibration. A single analytical observation produced a fitted shared stiffness scale of 0.9761. On 24 held-out geometries, MAPE decreased from 5.323% to 5.128%, while RMSE and MAE increased. A repeated-subset sensitivity analysis found that mean held-out MAPE remained worse than the nominal model for all tested calibration counts. The outcome is therefore mixed and remains preliminary. Earlier one-point compliance-multiplier and 100 N results are preserved separately for comparison.

1 Problem definition and scope

The engineering problem motivating the work is to reduce the cost of building response models for composite landing-gear geometry. Repeated physical testing can be expensive. The proposed future workflow is to generate an FEA response database, use a limited number of true observations to correct systematic model discrepancy, and predict responses at unobserved design points. This project does not include physical testing data. It studies whether the workflow can be implemented and tested against an analytical reference before application to curved landing gear.

The benchmark is a straight cantilever fixed at its root and loaded transversely at its free tip. The planform width decreases linearly from root width b_0 to tip width $r_b b_0$, where r_b is the taper ratio. The 20-ply sequence is fixed; cured ply thickness and effective elastic stiffness are swept. This benchmark tests a method: it is not a geometric representation of an undercarriage and does not model impact, contact, damage, ply failure, material nonlinearity, or dynamic landing response. The project scope and proposal require coordinator approval under the course brief; record the approval decision here before final submission: [**student: enter approval status and date**].

2 Inputs, outputs, and design domain

The fitted response is tip compliance $C = |\delta|/P$ in metres per newton. Fitting compliance makes sense for the linear elastic model because deflection scales with applied load. The current design inputs are beam length L, root width b_0, taper ratio r_b, total laminate thickness h, and a shared effective stiffness scale s_E. The nominal geometry, factor ranges, and 10 N load are defined in `config.quadratic.example.json`;

the 20-ply material sequence is defined in `layup.json`. These sweep ranges are a methods domain, not certified engineering limits.

Table 1: Inputs and domain specified for the current PyAnsys sweep.

Input/output	Symbol	Units	Values / domain
Beam length	L	m	0.40–0.60
Root width	b_0	m	0.040–0.060
Taper ratio	$r_b = b_{tip}/b_0$	–	0.40–0.90
Tip load	P	N	10 (fixed)
Layup thickness	h	m	0.002625–0.004375
Effective stiffness scale	s_E	–	0.80–1.20
Tip deflection	δ	m	FEA output; signed in CSV
Compliance	$C = \lvert\delta\rvert/P$	m/N	Fitting target
Laminate mass	m	kg	FEA post-processing output

The current fit requires finite values for all five inputs, the derived laminate section properties, and positive compliance. It uses a 120-case design; the 21-term quadratic surface is fitted to 96 training cases, with 24 held out. The laminate comprises ten 0° carbon twill plies and ten 0° glass plies, alternated through thickness. Cured ply thicknesses are 0.20 mm and 0.15 mm respectively, giving 3.50 mm nominal thickness before the thickness scale is applied. The glass is assumed to be UD E-glass/epoxy. The material constants are representative numerical inputs, not measured values; their sources and assumptions are recorded in `layup.json` and the bibliography. The alternating sequence is unsymmetric and may exhibit extension–bending coupling.

The scripts write CSV sweep data and prediction tables, JSON metrics, a serialized model artifact, and a LaTeX results table. MAPDL must be installed and licensed for solver runs. The fitting and plotting scripts operate on saved data and do not require MAPDL.

3 Analytical benchmark

For a small-deflection, linearly elastic beam with bending and Timoshenko shear deformation, tip compliance is evaluated as

$$C(L, b_0, r_b) = \int_0^L \left[\frac{(L-x)^2}{b(x) D_{eff}} + \frac{1}{\kappa b(x) S_{eff}} \right] dx, \tag{1}$$

$$b(x) = b_0 \left[1 - (1 - r_b) \frac{x}{L} \right]. \tag{2}$$

Here D_{eff} is laminate bending rigidity per unit width (N m), S_{eff} is transverse shear rigidity per unit width (N/m), and $\kappa = 5/6$ is the assumed shear correction factor. The reference value is $\delta = PC$. In the implementation, a 64-point Gauss–Legendre rule evaluates the integral.

For the 0° laminate, the code assembles the Classical Lamination Theory A, B, and D matrices from ply in-plane reduced stiffnesses, following standard laminate theory [4]. It obtains effective bending rigidity by inverting the full ABD matrix under zero membrane resultants and zero non-x bending moments, retaining coupling for the unsymmetric stack. The approximate shear rigidity is $S_{xz} = \sum_k G_{xz,k} t_k$. These are explicit benchmark choices; the shear model and material properties should be treated as assumptions. The comparison tests agreement between FEA and this chosen reference implementation, not truth for a real composite.

4 FEA and surrogate method

The current input configures a structured SHELL181 model with 40 elements along the beam, 6 across its width, and three through-thickness integration points per ply. SHELL181 supports layered sections defined through section data commands [5, 6]. Each case uses static linear analysis, a 10 N transverse tip-edge force, a clamped root, and 20 explicit plies. The runner writes a CSV containing geometry, total thickness, effective stiffnesses, deflection, compliance, mass, and mesh settings. Record the MAPDL release, PyMAPDL release, solver log, and run configuration when executing the sweep.

The model inputs are (L, b_0, r_b, h, s_E) and the output is C. A fixed random seed reserves 20% of cases for a held-out test set. The raw FEA response is fitted with a full degree-two polynomial surface with ridge regularization. Let $C_Q(\mathbf{x})$ be its prediction and $C_A(\mathbf{x})$ the analytical benchmark. Calibration estimates the effective stiffness scale directly by bounded least squares, using selected training geometries at a common thickness h_c:

$$\widehat{s}_E = \arg \min_{s_E \in [s_{min}, s_{max}]} \sum_{i \in \mathcal{C}} \left[C_A(L_i, b_i, r_i, h_c; s_{E,true}) - C_Q(L_i, b_i, r_i, h_c; s_E) \right]^2 . \tag{3}$$

The fitted scale multiplies all elastic and shear moduli, so bending and transverse shear stiffness change together. One calibration configuration is the mathematical minimum for one scalar; the default selects training geometries near the centre of the geometry domain. The held-out geometries are predicted at the same h_c and $\widehat{s}_E$ and compared with the analytical equation evaluated at the reference scale $s_{E,true}$. This isolates geometry transfer in the analytical comparison. A larger calibration set can estimate the same scalar more robustly. The nominal and reference scales are set in the input configuration; both equal 1.0 for this benchmark.

5 Results and validation

The revised direct-stiffness fit and its outputs are archived in `results/quadratic_run_2026-10-17_direct_stiffness_calibration/`. The 120-case dataset was divided into 96 training cases and 24 held-out cases. One analytical calibration observation (case 109) was used to estimate the shared effective stiffness scale, yielding $\widehat{s}_E = 0.97609$ relative to the reference scale of 1.0. The held-out geometry test keeps thickness at its nominal scale and compares against the analytical model at the reference stiffness.

Table 2: Held-out performance from the direct-stiffness calibration run.

Comparison	RMSE (m/N)	MAE (m/N)	R^2	MAPE (%)
Quadratic vs. FEA at DOE inputs	3.624×10^{-4}	2.712×10^{-4}	0.9876	5.929
Nominal stiffness vs. analytical	2.668×10^{-4}	2.246×10^{-4}	0.9777	5.323
Calibrated stiffness vs. analytical	3.040×10^{-4}	2.518×10^{-4}	0.9710	5.128

The calibration gives a mixed outcome. It reduces MAPE by 0.195 percentage points and reduces maximum absolute percentage error from 15.36% to 12.17%, but RMSE rises by 13.9%, MAE rises by 12.1%, and R^2 falls from 0.9777 to 0.9710. This shows that the fitted scalar improves the percentage-error measures for this held-out split, while worsening absolute-error measures. It does not establish a general improvement. One observation is the mathematical minimum for one fitted scalar; repeat the assessment with multiple calibration observations and alternative held-out splits to test robustness.

5.1 Sensitivity to the number and selection of observations

To test whether the result depends on calibration-case selection, the quadratic model and 24-case held-out geometry set were held fixed. For each calibration count (1, 2, 3, or 5), 30 unique calibration subsets

were evaluated: one centred/maximin subset and 29 random subsets sampled from the training cases. Analytical references for held-out cases were used only for scoring.

Table 3: Calibration-subset sensitivity on one fixed train/test split. Positive error changes mean worse performance than the nominal model.

Observations	Mean fitted s_E (SD)	Mean ΔMAPE (pp)	Median ΔMAPE (pp)	MAPE improved
1	0.9818 (0.0462)	+1.521	+0.919	30.0%
2	0.9978 (0.0307)	+0.830	+0.579	33.3%
3	1.0030 (0.0289)	+0.895	+0.692	30.0%
5	1.0055 (0.0164)	+0.535	+0.314	36.7%

The nominal held-out MAPE was 5.323%. Across these subsets, both mean and median MAPE change were positive at every calibration count, and only 30.0–36.7% of subsets improved MAPE. Mean RMSE increased by 53.0%, 22.8%, 20.5%, and 8.4% for 1, 2, 3, and 5 observations respectively. Increasing the number of observations made the fitted stiffness scale less variable and brought its mean closer to the reference value, but that did not produce better predictions on average for this split. Calibration-case placement matters, and the centred one-point result is not representative of most tested subsets.

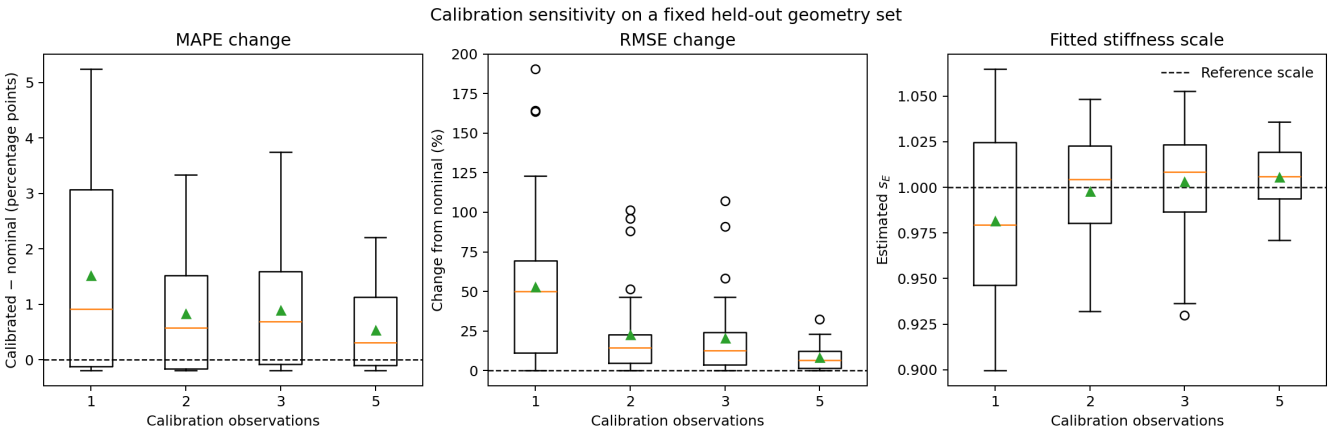


Figure 1: Held-out error changes and fitted stiffness scales across 30 calibration subsets per observation count. Dashed lines mark no error change and the reference stiffness scale.

The sensitivity analysis conditions on one fitted response surface and one train/test split. It tests calibration-subset sensitivity, not variation across different FEA designs or physical observations. Repeat with alternative fixed split seeds before making a broader claim.

The sweep's $|\delta|/L$ ratio ranges from approximately 0.034 to 0.330. Cases near the upper end may violate the small-deflection assumption of the analytical benchmark. The comparison tests interpolation within the sampled parameter region under stated assumptions. It does not establish extrapolation capacity or validate physical material properties. No physical test data are included, and no physical validation claim is made.

6 Validity, limitations, and next work

The 10 N load is chosen to reduce deflection relative to the archived exploratory case, but this does not establish small-deflection validity across the new thickness and stiffness range. Inspect $|\delta|/L$ for all cases and confirm compliance is independent of load in representative linear analyses before using the analytical comparison.

The archive has only one mesh resolution, so mesh convergence remains untested. Re-run representative cases with progressively refined longitudinal and width divisions and report changes in compliance. Record MAPDL and PyMAPDL versions, solver output, element formulation and settings, boundary conditions, and run configuration. Review the shell local axes and material orientation, supports, force application, stress singularities, shear locking risk, and section behavior. Run the automated Python tests; they do not test a MAPDL solution. The material constants and density assumptions are not a calibrated or measured property set. The model omits geometric nonlinearity, contact, failure, impact, rate effects, and dynamic landing behavior.

Evaluate interpolation with held-out RMSE, MAE, R^2, MAPE, and maximum relative error. Test extrapolation separately using a deliberately separated region beyond the training envelope. Repeat the split or use cross-validation to assess sensitivity to the design sample. Report runtime for FEA, fitting, and prediction using a stated machine and software version. A mass-optimal configuration can then be solved subject to a supplied deflection requirement, geometric bounds, and manufacturability constraints. Extending to a curved gear requires a new geometry parameterization and renewed validation; benchmark performance does not transfer automatically.

7 Brief alignment and project deliverables

The repository includes the functional source scripts, example inputs, results, unit-aware problem definition, a license, tests, test evidence, HTML API documentation, and this report draft. A requirement-by-requirement status and outstanding actions are listed in `docs/requirements_traceability.md`. The project brief also requires a short presentation; it should present the problem, method, results, validation limits, alternatives, performance evidence, GitHub repository, and AI disclosure. Before submission, ensure the final compiled report is within the 5–10 page limit and replace all explicit student action notes with reviewed, supported information.

8 AI use and student reflection

AI tools were used to assist with drafting and editing project code, generating documentation scaffolding, identifying relevant material-property sources, debugging, and presenting analysis and plots. The student remains responsible for checking the code, the analytical derivation, source applicability, units, solver setup, and all reported values. Generated text or code is not evidence of correctness.

Student reflection required: *[Add a concise first-person reflection describing the AI assistance actually used, one useful contribution, one limitation or error caught during review, how outputs were verified, and what decisions remained the student's own. Do not submit this placeholder unchanged.]*

References

[1] Gurit, *SE84LV High Performance Prepreg System: Product Data Sheet*, 2024. Available at `https://www.revchem.com/document/15pc135966a_tds-pdf/`.

[2] Performance Composites, *Mechanical Properties of Carbon Fibre Composite Materials*. Available at `https://performance-composites.com/carbonfibre/mechanicalproperties_2.asp`.

[3] University of South Florida, *Hooke's Law for a Two-Dimensional Unidirectional Lamina: Worked Example*. Available at `https://kaw.eng.usf.edu/compositesOCW/WorkedOutExamples/Section%202.4%20Hooke's%20Law%20for%20a%20Two%20Dimensional%20Unidirectional%20Lamina/Section2_4_worked_out_example_PPT/Section2_4_worked_out_example_PPT.html`.

[4] J. N. Reddy, *Mechanics of Laminated Composite Plates and Shells: Theory and Analysis*, 2nd ed., CRC Press, 2004.

[5] Ansys, Inc., *SHELL181 Element Documentation*. Available at `https://ansyshelp.ansys.com/public/account/secured?returnurl=/Views/Secured/corp/v242/en/ans_elem/Hlp_E_SHELL181.html`.

[6] Ansys, Inc., *PyMAPDL secdata API*. Available at `https://mapdl.docs.pyansys.com/version/stable/api/_autosummary/ansys.mapdl.core.Mapdl.secdata.html`.